

WITTENBERG UNIVERSITY

Mini SIEM:

Multi-Source Log Correlation and Behavioral Detection System

Senior Capstone Project Technical Report

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Abstract

The increasing volume and complexity of cybersecurity threats have created a growing need for Security Information and Event Management (SIEM) systems capable of aggregating and analyzing security data from multiple sources. Traditional enterprise SIEM platforms such as Splunk and Microsoft Sentinel provide centralized monitoring and threat detection but often require extensive infrastructure and licensing costs. This project explores the development of a simplified, lightweight SIEM platform capable of ingesting, correlating, detecting, and visualizing cybersecurity activity using real-world datasets.

The developed system utilizes the Los Alamos National Laboratory (LANL) cybersecurity dataset, including authentication, process, network flow, DNS, and red team activity logs. Custom Python ingestion scripts were developed to parse and normalize multiple log formats into a unified event schema, allowing events from different sources to be analyzed consistently. The normalized events were stored in a centralized SQLite database to support correlation and detection operations.

Behavior-based detection rules were implemented to identify suspicious activity patterns using multi-source correlation and time-based filtering. Detected activity was validated against LANL red team ground truth data to measure detection effectiveness. The system generated 16,732 structured alerts, with 12,445 associated with systems involved in known attacker activity, resulting in a validation rate of 74.38%.

To improve usability, a Flask-based dashboard was developed to visualize detection results, alert severity distributions, recent alert activity, and interactive filtering capabilities. The final system demonstrates the core functionality of a modern SIEM platform, including log ingestion, normalization, event correlation, behavioral detection, alert validation, and visualization in a lightweight and accessible implementation.

Introduction

Modern organizations generate massive amounts of system and network activity logs on a daily basis. Authentication attempts, process execution, network communication, and DNS activity all contribute to a large and continuously growing volume of cybersecurity data. Security teams rely on Security Information and Event Management (SIEM) systems to centralize, analyze, and monitor these logs for indicators of malicious behavior and potential security threats.

SIEM platforms play an essential role in cybersecurity by aggregating security data from multiple sources and applying correlation and detection techniques to identify suspicious activity. Enterprise SIEM solutions such as Splunk, Elastic Security (ELK Stack), and Microsoft Sentinel are commonly used to support threat detection, incident response, and security monitoring. However, many enterprise solutions require significant infrastructure, configuration, and licensing costs, making them difficult to fully understand or implement in smaller environments.

The purpose of this project is to design and implement a simplified Security Information and Event Management (SIEM) platform capable of performing core cybersecurity monitoring functions using real-world log data. The system focuses on ingesting and normalizing multiple log sources, storing events in a centralized database, correlating activity across datasets, generating behavior-based detections, validating results against known attacker activity, and visualizing alerts through a user-facing dashboard.

To accomplish this objective, the project utilizes the Los Alamos National Laboratory (LANL) cybersecurity dataset, which contains enterprise-style authentication, process, network flow, DNS, and red team activity logs. Unlike highly structured datasets that explicitly label attacks, the LANL dataset primarily consists of raw behavioral activity. This creates a more realistic cybersecurity scenario in which suspicious activity must be inferred through event relationships, timing, and validation techniques.

The implemented system applies multi-source event correlation and time-based behavioral detection logic to identify suspicious activity patterns. Detection results are validated against red team ground truth data to measure effectiveness. Additionally, a Flask-based dashboard provides visualization capabilities to support alert monitoring and investigation workflows.

The primary objectives of this project include:

- Develop a lightweight SIEM platform capable of ingesting multiple cybersecurity log sources
- Normalize log data into a unified event schema for consistent analysis
- Store events within a centralized SQLite database
- Correlate activity across authentication, process, network, and DNS logs
- Implement behavior-based detection rules for suspicious activity identification
- Validate detections using LANL red team data
- Develop an interactive dashboard for alert visualization and monitoring

This project demonstrates how the core functionality of a modern SIEM system can be implemented using open-source tools and real-world cybersecurity data while maintaining accessibility and scalability for educational and research purposes.

Literature Review

Security Information and Event Management (SIEM) systems have become a critical component of modern cybersecurity operations. As organizations generate increasingly large volumes of system, network, and user activity logs, centralized monitoring platforms are necessary to aggregate, analyze, and correlate security events in order to identify potential threats. SIEM platforms provide security analysts with visibility into enterprise activity while supporting incident detection, investigation, and response.

According to Splunk, a SIEM system combines Security Information Management (SIM) and Security Event Management (SEM) capabilities to centralize log collection and automate threat detection through event correlation and analytics. Modern SIEM platforms ingest data from authentication systems, endpoints, applications, firewalls, intrusion detection systems, and network devices to create a unified security monitoring environment.

One of the primary advantages of SIEM systems is their ability to correlate activity across multiple data sources. Correlation refers to the process of identifying relationships between events that may individually appear benign but collectively indicate suspicious behavior. For example, an authentication event followed closely by process execution and network communication may suggest post-compromise activity or lateral movement within a network environment.

Research in cybersecurity increasingly emphasizes behavioral detection rather than signature-based detection methods. Traditional detection systems often rely on predefined attack signatures or known indicators of compromise, which may fail to identify novel threats. Behavioral detection instead focuses on identifying abnormal activity patterns and suspicious event relationships. According to Scarfone and Mell (2007), log analysis and event correlation are fundamental techniques used to improve situational awareness and threat detection within enterprise security environments.

Enterprise SIEM platforms such as Splunk, Elastic Security (ELK Stack), IBM QRadar, and Microsoft Sentinel provide sophisticated correlation, alerting, and visualization capabilities. Splunk supports centralized log analysis and customizable search functionality for threat detection, while the Elastic Stack emphasizes scalable log ingestion and visualization using Elasticsearch and Kibana. Microsoft Sentinel provides cloud-native SIEM functionality using Microsoft Azure services and automated security analytics.

Despite their advantages, enterprise SIEM systems often present challenges including infrastructure complexity, licensing costs, configuration overhead, and operational resource requirements. Smaller organizations and educational environments may not have access to enterprise-scale security monitoring solutions. As a result, lightweight SIEM

implementations have become increasingly valuable for cybersecurity research, education, and proof-of-concept development.

The Los Alamos National Laboratory (LANL) cybersecurity dataset has become widely used within cybersecurity research for evaluating detection and correlation methodologies. The dataset includes enterprise-scale authentication logs, process execution logs, network flow activity, DNS activity, and red team attacker movement data. Unlike datasets that explicitly label attacks, the LANL dataset contains mostly raw behavioral activity, requiring analysts to infer suspicious behavior through correlation and contextual analysis. This characteristic makes the dataset particularly valuable for studying behavioral detection strategies and realistic cybersecurity monitoring scenarios.

Several studies have highlighted the importance of time-based correlation in cybersecurity monitoring. Temporal relationships between events often improve detection accuracy by reducing false correlations between unrelated activities. Events occurring within short time windows may indicate automated or coordinated attacker behavior, while broader event windows can introduce additional noise and false positives.

Visualization also plays a significant role in SIEM effectiveness. Dashboards and monitoring interfaces enable analysts to quickly identify trends, prioritize alerts, and investigate suspicious systems. According to Kent (2015), visualization improves analyst efficiency by presenting complex security information in a structured and accessible manner.

This project builds upon existing SIEM concepts by combining multi-source log ingestion, event normalization, time-based behavioral correlation, detection engineering, alert validation, and dashboard visualization into a lightweight implementation. While simplified compared to enterprise solutions, the system demonstrates many of the core functions performed by modern SIEM platforms.

Requirements and Specifications

The Mini SIEM platform was designed to meet several functional requirements necessary for cybersecurity monitoring and behavioral threat detection. Functional requirements define the actions and capabilities that the system must perform.

The implemented functional requirements include:

Log Ingestion

The system must ingest multiple cybersecurity log sources, including authentication, process, network flow, DNS, and attacker activity logs.

Event Normalization

The system must normalize multiple log formats into a unified event schema to ensure consistent analysis across datasets.

Event Storage

The system must store normalized events within a centralized database to support querying, correlation, and alert generation.

Event Correlation

The system must identify relationships between authentication, process, network, and DNS activity across shared systems and timestamps.

Behavioral Detection

The system must generate alerts based on suspicious behavioral patterns identified through time-based correlation rules.

Alert Validation

The system must validate detections using LANL red team ground truth data.

Dashboard Visualization

The system must provide a user-facing dashboard capable of displaying alerts, metrics, severity levels, and filtering capabilities.

Non-Functional Requirements

In addition to functional requirements, several non-functional requirements were considered during system design and implementation.

Performance

The system should efficiently process large cybersecurity datasets while maintaining reasonable performance during querying and visualization.

Scalability

The architecture should support future expansion to larger datasets or additional log sources.

Maintainability

The system should use modular Python scripts to simplify updates and improvements.

Usability

The dashboard interface should remain readable, intuitive, and accessible for users with cybersecurity knowledge.

Reliability

The system should consistently normalize, correlate, and store event data without data loss or processing failures.

Security

The platform should use trusted research datasets and maintain responsible cybersecurity practices throughout development.

Software

Several software tools and frameworks were used throughout the development of the Mini SIEM platform.

Software	Purpose
Python	Core programming language used for ingestion, normalization, correlation, and detection
Visual Studio Code	Primary development environment
Flask	Web framework used to build the dashboard
SQLite	Lightweight relational database used for event and alert storage
JSON / JSONL	Structured event and alert storage format
Chart.js	Dashboard visualization framework
GitHub	Source control and version management
Canva	System architecture and presentation diagram creation
Windows 11	Development operating system

Hardware

The Mini SIEM platform was developed and tested using a local workstation environment. Since the project was designed as a lightweight proof-of-concept implementation, enterprise infrastructure was not required. All ingestion, normalization, correlation, detection, database operations, and dashboard visualization were performed locally.

The hardware specifications used for development include:

Hardware Component	Specification
Processor (CPU)	Intel Core Ultra 9 185H
Memory (RAM)	32 GB RAM
Storage	SSD Storage
Operating System	Windows 11

Development Environment Local machine execution

The hardware environment provided sufficient resources for processing LANL log samples, generating alerts, storing event data within SQLite, and running the Flask dashboard interface without significant performance limitations.

User

The Mini SIEM platform is designed for users who require visibility into cybersecurity activity and behavioral threat detection. While simplified compared to enterprise SIEM platforms, the system provides functionality useful for monitoring, research, and educational purposes.

The primary intended users include:

Security Analysts

Security analysts can use the system to monitor suspicious activity, review alerts, analyze behavioral patterns, and investigate affected systems through the dashboard interface.

Cybersecurity Students

Students studying cybersecurity can use the platform to better understand SIEM workflows, event correlation, detection engineering, and security monitoring concepts.

Researchers

Researchers can utilize the system to experiment with detection rules, behavioral analytics, and log correlation techniques using realistic enterprise datasets.

IT and Security Administrators

Administrators may use similar concepts for monitoring authentication, process, and network activity within smaller environments.

The dashboard interface was intentionally designed to remain simple and intuitive while still providing meaningful cybersecurity monitoring functionality.

System Design

The Mini SIEM platform was designed using a modular architecture to separate ingestion, normalization, storage, detection, and visualization components into manageable layers. This design improves maintainability, scalability, and readability while allowing individual system components to be modified independently.

The overall design consists of the following major components:

Ingestion Layer

The ingestion layer processes raw LANL cybersecurity logs. Custom Python scripts were developed for each log source to read and parse authentication, process, network flow, DNS, and red team activity logs.

Normalization Layer

The normalization layer converts raw log entries into a unified event schema. Standardized fields such as timestamps, event types, users, systems, and metadata are used to ensure consistency across datasets.

Database Layer

Normalized events are stored within a centralized SQLite database. This layer enables querying, event persistence, and support for multi-source correlation and detection logic.

Correlation Layer

The correlation engine identifies relationships between multiple log sources based on shared systems, users, and timestamps. Time-based filtering improves detection realism and reduces unrelated event matches.

Detection Layer

Behavior-based detection rules analyze correlated activity to identify suspicious event chains involving authentication, process execution, network communication, and DNS activity.

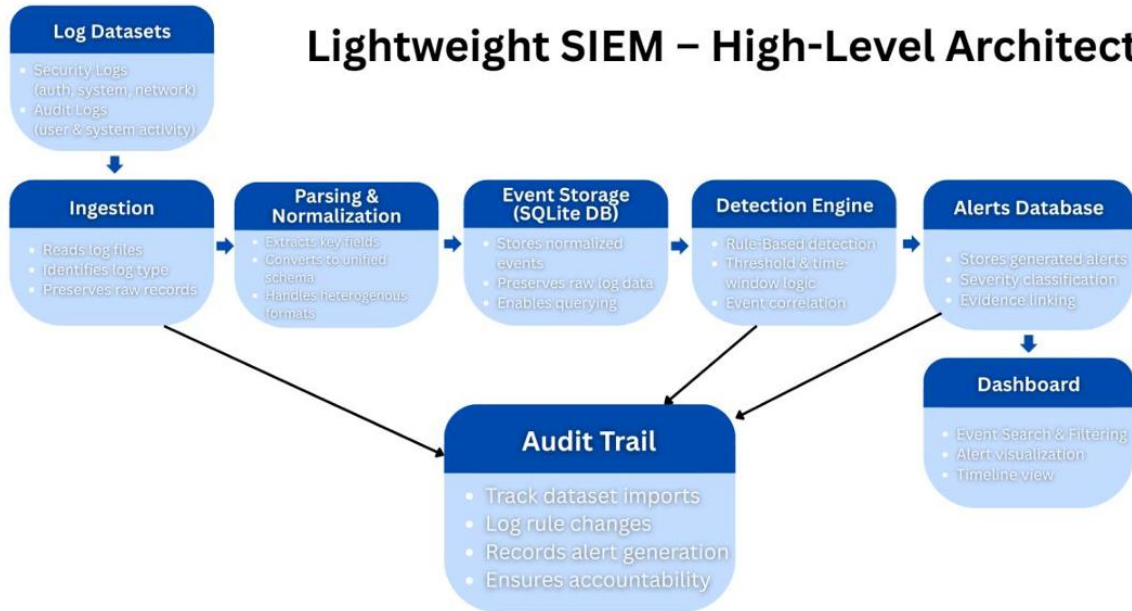
Visualization Layer

A Flask-based dashboard provides alert visualization, severity monitoring, rule distribution charts, recent alerts, and interactive filtering capabilities.

The modular design allows each system component to function independently while contributing to the overall SIEM workflow.

System Workflow

Lightweight SIEM – High-Level Architecture



Class Diagram

The Mini SIEM platform utilizes several core logical components that work together to process events, generate detections, and display monitoring results. While the system is implemented primarily through modular Python scripts rather than object-oriented software architecture, the following conceptual classes represent the major system components.

Major System Components

NormalizedEvent

Responsible for storing standardized event data including timestamps, event type, source, system information, and metadata.

Alert

Represents generated detections, including severity level, rule name, affected systems, timestamps, and descriptive information.

DatabaseManager

Handles SQLite database interactions including event insertion, alert storage, and querying.

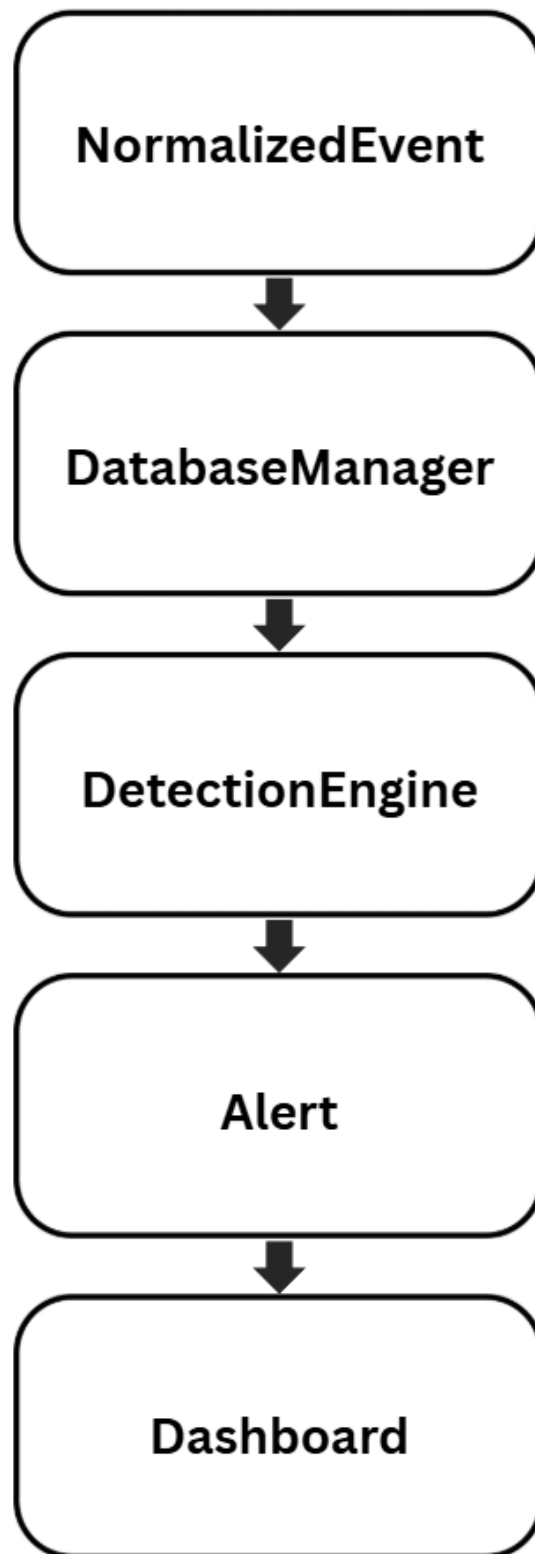
DetectionEngine

Applies behavior-based correlation rules to identify suspicious activity patterns.

Dashboard

Displays alerts, metrics, charts, and filtering functionality through the Flask user interface.

Conceptual class diagram of Mini SIEM components



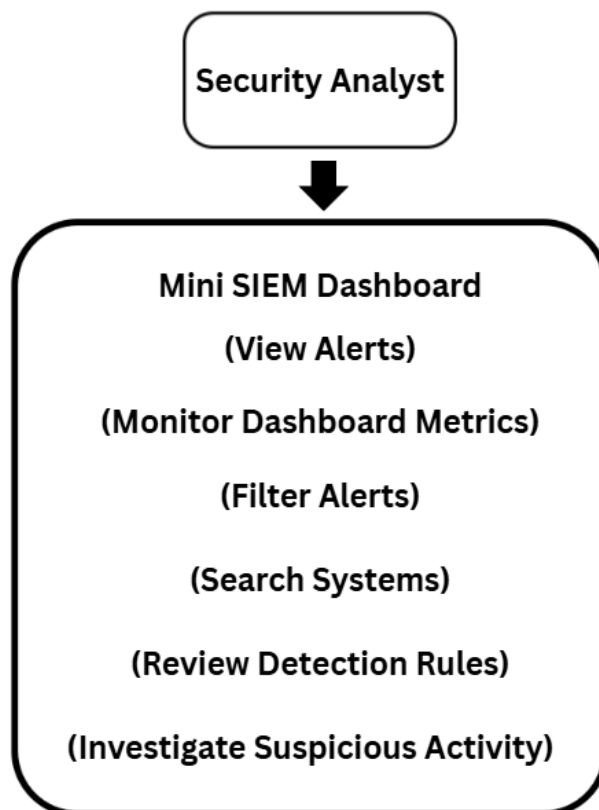
Use Case Diagram

The Mini SIEM platform was designed around a primary user role: the Security Analyst. The analyst interacts with the dashboard to monitor alerts, review suspicious activity, and investigate affected systems.

The major system use cases include:

- View system metrics
- Monitor generated alerts
- Review severity classifications
- Filter alerts by severity or rule
- Search affected systems
- Review detection rule information
- Investigate suspicious activity patterns

Mini SIEM use case diagram



Architecture / Model

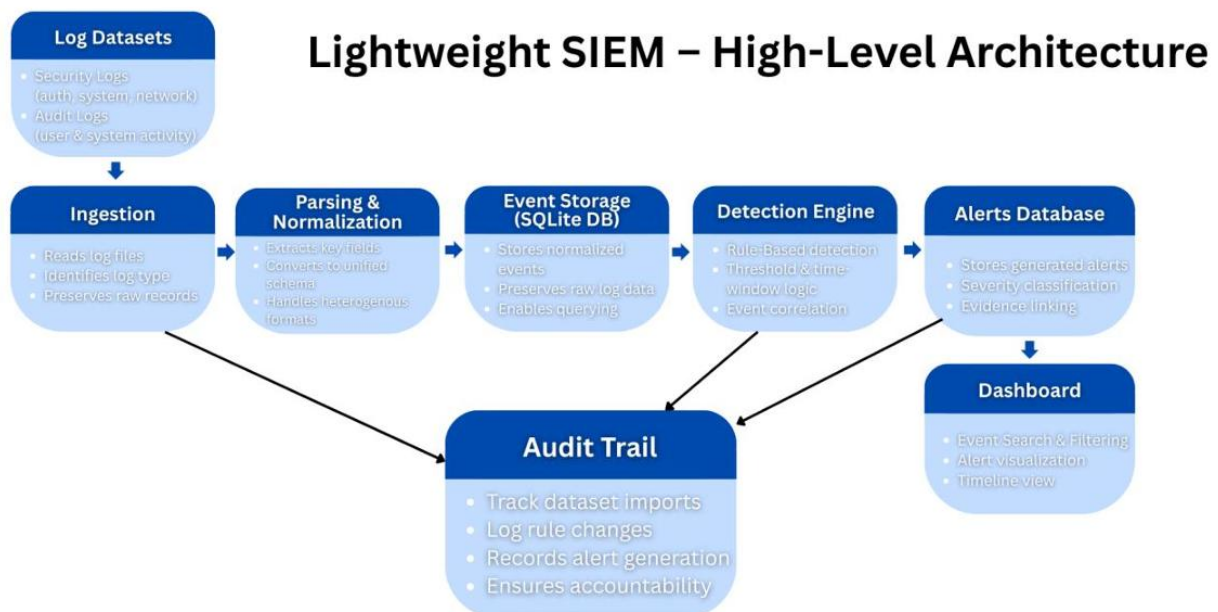
The Mini SIEM platform follows a layered architecture model designed to separate data ingestion, event processing, storage, detection, and visualization into independent system layers.

The architecture begins with multiple LANL cybersecurity log sources including authentication, process, network flow, DNS, and red team datasets. These logs are processed through custom Python ingestion scripts and normalized into a unified event schema.

The normalized events are stored within SQLite to support centralized querying and persistence. A correlation engine identifies relationships across datasets using system identifiers and timestamps. Detection rules are then applied to generate structured alerts based on suspicious behavior patterns.

Finally, a Flask-based dashboard visualizes the resulting alerts and system metrics while supporting filtering and investigation capabilities.

Layered architecture model of the Mini SIEM platform



Implementation and Source Control

The Mini SIEM platform was implemented primarily using Python due to its flexibility, readability, and strong support for cybersecurity and data processing workflows. The implementation process followed a modular design approach in which separate components were developed for ingestion, normalization, storage, correlation, detection, and visualization.

Log Ingestion Implementation

Custom Python scripts were created to process each LANL log source independently. Dedicated ingestion scripts were developed for:

- authentication logs (auth.txt)
- process execution logs (proc.txt)
- network flow logs (flows.txt)
- DNS activity logs (dns.txt)
- red team attacker activity logs (redteam.txt)

Each ingestion script parses raw log entries and forwards the data to the normalization layer.

Event Normalization Implementation

Raw log formats differed significantly across LANL sources. To support correlation and detection, custom normalization scripts were developed to convert all events into a unified schema containing standardized fields such as:

- timestamp
- event type
- user information
- source system
- destination system
- metadata

This standardized structure allowed events from multiple datasets to be analyzed consistently.

Database Implementation

SQLite was selected as the primary storage solution due to its lightweight design and ease of integration with Python. The database stores:

- normalized events
- detection alerts
- correlation data

SQLite enabled centralized querying and efficient retrieval of event information for both detection and dashboard visualization.

Correlation and Detection Implementation

The correlation engine was implemented using shared system identifiers and timestamps to identify relationships across multiple log sources.

Behavior-based detection rules were implemented using time-based filtering to identify suspicious activity chains such as:

- authentication followed by process execution
- process execution followed by network communication
- authentication → process → network activity chains
- DNS activity associated with attacker systems

The detection engine generated structured alerts that included severity classifications, timestamps, affected systems, and descriptive information.

Dashboard Implementation

A Flask-based dashboard was developed to provide a user interface for monitoring detection results. The dashboard dynamically retrieves data from SQLite and provides:

- alert metrics
- severity visualizations
- rule distributions
- recent alert monitoring
- interactive filtering capabilities

Source Control

GitHub was used for version management and source control throughout development. Source control provided backup, version tracking, and organizational support while enabling project files to be structured into modular components.

The final GitHub repository contains:

- ingestion scripts
- normalization scripts
- detection logic
- Flask dashboard implementation
- SQLite database structure
- supporting project documentation

Testing and Results

Extensive testing was performed throughout development to verify the correctness of ingestion, normalization, correlation, detection, and visualization functionality. Each system component was tested independently before being integrated into the complete Mini SIEM workflow.

Log Ingestion Testing

Each LANL dataset was tested independently to confirm that ingestion scripts correctly parsed raw logs and normalized events without data corruption or processing failures.

Successful ingestion was verified for:

- authentication logs
- process execution logs
- network flow logs
- DNS activity logs
- red team activity logs

Testing confirmed successful normalization and storage of approximately 20,000 sample records per major dataset.

Successful ingestion and normalization of LANL log sources

```
=== LANL Auth Integration Test ===  
Input file: data/raw/auth_targeted_sample.txt  
Normalized events written to: data/normalized/lanl_auth_events.jsonl  
Total events normalized: 20000  
Parse failures: 0
```

```
=== LANL Red Team Integration Test ===  
Input file: data/raw/redteam.txt  
Normalized events written to: data/normalized/lanl_redteam_events.jsonl  
Total events normalized: 749  
Parse failures: 0
```

```
=== LANL Process Integration Test ===  
Input file: data/raw/proc_sample.txt  
Normalized events written to: data/normalized/lanl_proc_events.jsonl  
Total events normalized: 20000  
Parse failures: 0
```

```
=== LANL Network Flow Integration Test ===  
Input file: data/raw/flows_sample.txt  
Normalized events written to: data/normalized/lanl_flows_events.jsonl  
Total events normalized: 20000  
Parse failures: 0
```

```
=== LANL DNS Integration Test ===  
Input file: data/raw/dns_sample.txt  
Normalized events written to: data/normalized/lanl_dns_events.jsonl  
Total events normalized: 20000  
Parse failures: 0
```

Correlation Engine Testing

The correlation engine was tested to verify that relationships between authentication, process, network flow, and DNS activity were successfully identified.

Testing confirmed overlap between red team systems and multiple event sources:

- Authentication matches: 49,818
- Process matches: 6,668
- Flow matches: 29,007
- DNS matches: 42,947

These overlaps demonstrated successful multi-source correlation across behavioral activity.

Multi-source correlation testing results

```
=== LANL Multi-Source Correlation Check ===  
  
Events by source:  
( 'lanl_auth', 20000)  
( 'lanl_dns', 20000)  
( 'lanl_flows', 20000)  
( 'lanl_proc', 20000)  
( 'lanl_redteam', 749)  
  
Redteam destination overlap:  
Auth matches: 49818  
Process matches: 6668  
Flow matches: 29007  
DNS matches: 42947
```

Detection Engine Testing

The behavior-based detection engine was tested to validate the effectiveness of implemented detection rules.

Final detection results include:

Metric	Result
Total Structured Alerts	16,732
Redteam-Validated Alerts	12,445
Validation Rate	74.38%

Detection rules successfully identified suspicious activity chains using time-based behavioral analysis and event correlation.

Implemented detection rules included:

- timebound_auth_process_chain
- timebound_process_network_chain
- auth_process_network_chain
- redteam_dns_overlap

Detection engine results and alert generation metrics

```
=== LANL Improved Detection Engine v2 ===
Time window threshold: 500
Limit per rule: 5000

Total structured alerts generated: 16732
Redteam-validated alerts: 12445
Validation rate: 74.38%

Alerts by rule:
  timebound_auth_process_chain: 1732
  timebound_process_network_chain: 5000
  auth_process_network_chain: 5000
  redteam_dns_overlap: 5000

Alerts by severity:
  high: 7445
  low: 4287
  critical: 5000
```

Dashboard Testing

The Flask dashboard was tested to confirm:

- successful SQLite connectivity
- correct alert retrieval
- chart rendering functionality
- severity visualization accuracy
- filtering functionality

Testing confirmed successful display of:

- alert metrics
- severity charts
- rule distributions
- recent alerts
- interactive filtering

Mini SIEM dashboard visualization

Mini SIEM Dashboard

LANL Multi-Source Detection and Validation Dashboard

80749

Total Events

16732

Total Alerts

12445

Redteam-Validated Alerts

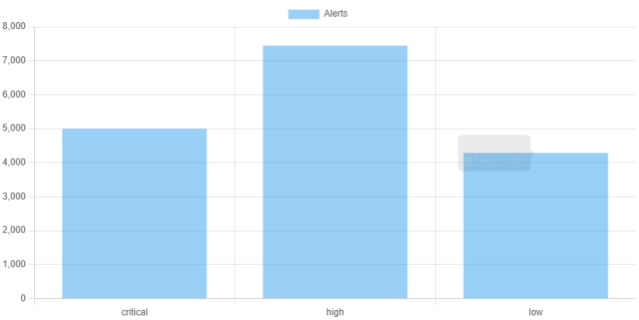
5000

Critical Alerts

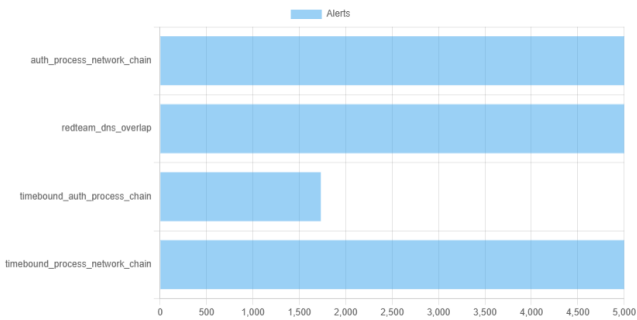
74.38%

Validation Rate

Alerts by Severity



Alerts by Rule



Alert Filters

All Severities All Rules Search affected system...

Recent Alerts

Timestamp	Rule	Severity	Affected System	Description
2026-05-11T02:28:43.467367+00:00	redteam_dns_overlap	HIGH	C11336	Redteam target C359 correlated with DNS activity involving C359.
2026-05-11T02:28:43.467324+00:00	redteam_dns_overlap	HIGH	C9063	Redteam target C359 correlated with DNS activity involving C359.
2026-05-11T02:28:43.467322+00:00	redteam_dns_overlap	HIGH	C154	Redteam target C359 correlated with DNS activity involving C359.
2026-05-11T02:28:43.467319+00:00	redteam_dns_overlap	HIGH	C5948	Redteam target C359 correlated with DNS activity involving C359.
2026-05-11T02:28:43.467316+00:00	redteam_dns_overlap	HIGH	C5857	Redteam target C359 correlated with DNS activity involving C359.
2026-05-11T02:28:43.467314+00:00	redteam_dns_overlap	HIGH	C8483	Redteam target C359 correlated with DNS activity involving C359.
2026-05-11T02:28:43.467311+00:00	redteam_dns_overlap	HIGH	C6509	Redteam target C359 correlated with DNS activity involving C359.
2026-05-11T02:28:43.467308+00:00	redteam_dns_overlap	HIGH	C6810	Redteam target C359 correlated with DNS activity involving C359.
2026-05-11T02:28:43.467306+00:00	redteam_dns_overlap	HIGH	C4600	Redteam target C359 correlated with DNS activity involving C359.
2026-05-11T02:28:43.467303+00:00	redteam_dns_overlap	HIGH	C9578	Redteam target C359 correlated with DNS activity involving C359.
2026-05-11T02:28:43.467300+00:00	redteam_dns_overlap	HIGH	C8951	Redteam target C359 correlated with DNS activity involving C359.
2026-05-11T02:28:43.467297+00:00	redteam_dns_overlap	HIGH	C2850	Redteam target C359 correlated with DNS activity involving C359.
2026-05-11T02:28:43.467294+00:00	redteam_dns_overlap	HIGH	C6346	Redteam target C359 correlated with DNS activity involving C359.
2026-05-11T02:28:43.467292+00:00	redteam_dns_overlap	HIGH	C6346	Redteam target C359 correlated with DNS activity involving C359.

Overall Results

Testing confirmed that the Mini SIEM platform successfully performs the core functions of a Security Information and Event Management system, including:

- multi-source log ingestion
- event normalization
- centralized storage
- behavioral correlation
- time-based detection
- alert generation
- validation using ground truth attacker data
- dashboard visualization

The final system demonstrated strong performance within a lightweight local environment while successfully identifying suspicious behavioral activity.

Ethical, Legal, and Security Considerations

Several ethical, legal, and security considerations were evaluated during the design and implementation of the Mini SIEM platform.

Ethical Considerations

The project was conducted exclusively using publicly available cybersecurity research datasets from Los Alamos National Laboratory. No unauthorized access, penetration testing, or collection of private user information occurred during development.

Behavior-based detection systems may generate false positives, meaning legitimate user activity can occasionally appear suspicious. Because of this limitation, alerts should be interpreted as indicators requiring investigation rather than definitive proof of malicious intent.

The project also emphasizes responsible cybersecurity practices and ethical detection engineering by focusing on defensive monitoring rather than offensive activity.

Legal Considerations

The LANL dataset used throughout development is publicly available for cybersecurity research and educational purposes. All software tools used within the project comply with open-source licensing or approved educational use.

No sensitive organizational logs, personal information, or proprietary datasets were collected or processed.

Security Considerations

Because the system operates locally using static datasets, live network security risks are minimal. However, if expanded into a real-world implementation, additional security measures would be required, including:

- user authentication
- access controls
- encrypted communications
- secure log storage
- role-based authorization

Future implementations would also require protections against unauthorized dashboard access and alert tampering.

Project Performance, Uniqueness, and Similarity to Others

The Mini SIEM platform demonstrated strong performance as a lightweight cybersecurity monitoring system capable of processing, correlating, detecting, and visualizing suspicious activity using realistic enterprise-style datasets.

Project Performance

The system successfully ingested and normalized approximately 20,000 sample records across multiple LANL log sources while maintaining reliable processing and database performance within a local execution environment.

The correlation engine successfully identified relationships across authentication, process, network flow, and DNS activity, allowing the reconstruction of behavioral activity patterns across systems.

The implemented detection engine generated:

- **16,732 structured alerts**
- **12,445 redteam-validated alerts**
- **74.38% validation rate**

These results demonstrate the effectiveness of time-based behavioral correlation and rule-based detection within the selected dataset.

Additionally, the Flask dashboard successfully visualized alert metrics, severity distributions, rule activity, recent alerts, and filtering functionality without major performance limitations.

Project Uniqueness

Several characteristics make this project unique compared to simplified cybersecurity monitoring systems.

First, the platform utilizes **multi-source enterprise-style behavioral data** rather than highly structured datasets with predefined attack labels. This creates a more realistic cybersecurity environment in which suspicious behavior must be inferred through event relationships and contextual analysis.

Second, the system incorporates **time-based behavioral correlation**, improving detection realism by reducing unrelated event matches and emphasizing meaningful behavioral chains.

Third, the project integrates **red team validation data** to evaluate detection effectiveness against known attacker activity. This provides measurable evidence of system performance rather than relying solely on theoretical detections.

Finally, the project combines:

- log ingestion
- normalization
- centralized storage
- behavioral detection
- validation
- dashboard visualization

within a single lightweight implementation.

Similarity to Enterprise SIEM Platforms

Although simplified for educational and research purposes, the Mini SIEM platform shares several similarities with enterprise SIEM systems such as Splunk, Microsoft Sentinel, IBM QRadar, and Elastic Security.

Similarities include:

- centralized log aggregation
- event normalization
- multi-source correlation
- behavioral detection logic
- alert generation
- dashboard visualization
- analyst investigation workflows

However, enterprise SIEM platforms generally include additional capabilities such as:

- real-time ingestion pipelines
- large-scale distributed storage
- advanced threat intelligence integration

- machine learning detection models
- automated incident response workflows
- user authentication and enterprise security controls

The Mini SIEM project demonstrates the core concepts behind these enterprise technologies while maintaining simplicity and accessibility for educational use.

Conclusion

The objective of this project was to design and implement a simplified Security Information and Event Management (SIEM) platform capable of ingesting, normalizing, correlating, detecting, validating, and visualizing cybersecurity activity using real-world datasets.

The final system successfully achieved these objectives through the implementation of a modular architecture consisting of log ingestion, event normalization, centralized storage, behavioral correlation, rule-based detection, and dashboard visualization components.

Using the Los Alamos National Laboratory cybersecurity dataset, the platform successfully processed authentication, process, network flow, DNS, and red team activity logs. Multi-source correlation and time-based behavioral detection logic were applied to identify suspicious activity patterns across systems.

The implemented detection engine generated **16,732 structured alerts**, of which **12,445 were associated with systems involved in known attacker activity**, resulting in a **validation rate of 74.38%**. These results demonstrate that meaningful cybersecurity detections can be achieved through behavioral analysis and event correlation, even when working with largely unlabeled datasets.

The addition of a Flask-based dashboard significantly improved usability by providing a centralized monitoring interface capable of visualizing alerts, severity classifications, rule distributions, and interactive filtering capabilities.

This project demonstrates how the core functionality of modern SIEM systems can be implemented using lightweight tools and open-source technologies while maintaining meaningful cybersecurity monitoring capabilities. Although simplified compared to enterprise solutions, the Mini SIEM platform successfully replicates many of the fundamental concepts used in real-world security operations.

Future improvements may include real-time log ingestion, machine learning-based anomaly detection, cloud integration, enhanced dashboard capabilities, and enterprise-scale deployment options.

Overall, this project provided valuable experience in cybersecurity engineering, behavioral threat detection, event correlation, and SIEM system design while demonstrating the practical application of cybersecurity concepts within a real-world research environment.

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